

Zero Divisors: Let R be a ring. An element $x \in R$ is called a zero-divisor if there is a nonzero $y (\neq 0)$ with $xy = 0$; otherwise, x is called a non zero divisor.

$Z.\text{div}(R) \equiv \{\text{set of zero divisors}\}$

$S_0 = \{\text{set of non zero divisors}\}$

* A subset $S \subset R$ is called multiplicative if

(i) $1 \in S$

(ii) $x, y \in S \Rightarrow xy \in S$

• S_0 is multiplicative

Def: An ideal $\mathfrak{p}$ is called prime if its complement $(R - \mathfrak{p})$ is multiplicative or if

(i) $1 \notin \mathfrak{p}$

(ii) if $xy \in \mathfrak{p} \Rightarrow x \in \mathfrak{p}$ or $y \in \mathfrak{p}$

Proof: Suppose (i): $(R - \mathfrak{p})$ is mult.

then $1 \in (R - \mathfrak{p})$ so (i) is satisfied

and so $x, y \in (R - \mathfrak{p}) \Rightarrow xy \in S$

$xy \in \mathfrak{p}$, then since $\mathfrak{p}$ is an ideal

we have x or $y \in \mathfrak{p}$. If they both

exist in $\mathfrak{p}$ this is trivial.

$\leadsto$ Recall, if $1 = 0$, then $\forall x \in R$,

$$x = x \cdot 1 = x \cdot 0 = 0$$

i.e. the only element in R is 0.

* A ring is called a field if $1 \neq 0$ and if every nonzero element is a unit.

ex: $\mathbb{Q}, \mathbb{R}, \mathbb{C}, \mathbb{F}_q$ (Domain)

Def: A ring R is called an integral domain if $1 \neq 0$ is prime, or equivalently,

R is nonzero and has no nonzero divisors

Trivially: $R \setminus \{0\}$ is multiplicative and hence prime.

R is non-zero since $\{0\}$ is prime and hence $1 \in R$.

It has no non-zero divisors since

$R \setminus \{0\}$ is prime and so

$$x, y \in (R \setminus \{0\}) \Rightarrow xy \in (R \setminus \{0\})$$

$$\text{or, } xy \neq 0, \quad x, y \neq 0$$

ex: $\mathbb{Z}$, $K[x_1, \dots, x_n]$

* Every domain R is a subring of its fraction field

$$\text{Frac}(R) = \left\{ \frac{x}{y} \mid x, y \in R, y \neq 0 \right\}$$

Conversely, any subring R of a field K , including K itself, is a domain.

- Any $x \neq 0 \in R$ cannot be a zero divisor since $xy = 0 \Rightarrow \frac{1}{x}(xy) = 0$, so $y = 0$.

Frac(R): has the CMP: the inclusion of R into any field L extends uniquely to an inclusion of $\text{Frac}(R)$ into L

ex: $\mathbb{Z}$ is a domain: $\mathbb{Z} \setminus \{0\}$ is multiplicative

Then,

$$\text{Frac}(\mathbb{Z}) = \mathbb{Q} \subset \mathbb{R} \subset \mathbb{C}$$

Polynomials over a domain: Let R be a domain

$X := \{X_\lambda\}_{\lambda \in \Lambda}$ a set of variables,

- Set $P := R[X]$. Then P is a domain as well.

$$F, G \in P \Rightarrow F = \sum_n a_{(i_1, \dots, i_n)} X^{(i_1, \dots, i_n)}$$

$$FG \neq 0$$

grlex leading term of FG is the product of the grlex leading terms of F and G

$$\deg(FG) = \deg(F) + \deg(G)$$

- * With a given ordering, order all monomials M of the same degree in the lexicographic order of exponents.
- Among the M on F with $\deg(M) = \deg(F)$
- the largest is called the grlex leading monomial of F
 - The grlex leading term is the product aM , $a \in R$
 a : grlex leading coefficient.
- * The grlex leading term of FG is the product of those aM and bN of F and G and so
- $$\deg(FG) = \deg(F) + \deg(G)$$
- proof: Since R is a domain, $a, b \in R$
 $a, b \neq 0$ and $ab \in R \neq 0$
- So $ab \neq 0$ and the product does not "vanish".
- $$\deg(MN) = \deg(M) + \deg(N)$$
- $\deg(MN) \geq \deg(M'N')$
for all pairs $M'N'$ in F and G
equality holds only if $\deg(M) = \deg(F)$
 $\deg(N) = \deg(G)$
- Thus $abMN$
is the grlex leading term of FG
- * Among the monomials M in F with $\text{ord}(M) = \text{ord}(F)$
the smallest is called the grlex hind monomial of F .
- grlex hind term of F is the product

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where $a \in R$ is the coeff. of M in F .

Claim: The glex lead term of FG is the prod of glex lead terms of F and G .

$(x_\lambda) \in R^\lambda$, then

$$\text{ord}_{(x_\lambda)}(FG) = \text{ord}_{(x_\lambda)} F + \text{ord}_{(x_\lambda)} G$$

• If $FG = 1$, then $F, G \in R$

Frac(R) is called the field of rational functions

$$\underline{K(x)}, \quad \underline{K := \text{Frac}(R)}$$

Unique factorization: Let R be a domain, p a non zero non unit:

def: We call p prime if, whenever $p \mid xy$

$$\left[\begin{array}{l} p \mid xy \equiv (\exists z \in R \text{ s.t. } pz = xy) \\ \text{either} \\ p \mid x \text{ or } p \mid y \end{array} \right]$$

• Clearly, p is prime iff the ideal $\langle p \rangle$ is prime.

proof:

($\Rightarrow$) Suppose p is prime, then $(R \setminus \langle p \rangle)$:

• Note $p \neq 1$ since, if $p = 1$ then $p \mid xy \Rightarrow \exists z \text{ s.t. } pz = xy$
or $z \cdot 1 = xy$

so $1 \mid x$ and $1 \mid y$
thus $\langle p \rangle \neq (1)$

• Suppose $xy \in \langle p \rangle$
then $\exists n \in \mathbb{Z}$ s.t.
 $p^n = xy$

$$p(p^{n-1}) = xy$$

then

$$p|x \quad \text{or} \quad p|y$$

$$\omega p = x \quad \text{or} \quad \gamma p = y$$

Since $\langle p \rangle$ is an ideal, then

$$x \in \langle p \rangle \quad \text{or} \quad y \in \langle p \rangle.$$

($\Leftarrow$) Suppose $\langle p \rangle$ is prime, or

$(R \setminus \langle p \rangle)$ is multiplicative

$$(R \setminus \langle p \rangle) \ni 1 \quad \text{so} \quad p \neq 1$$

$$x, y \in (R \setminus \langle p \rangle)$$

$$\Rightarrow \text{Suppose } x \cdot y = p^l$$

then ~~$p^n \in \langle p \rangle$~~

Since $x \cdot y \in \langle p \rangle$, then

$$x \in \langle p \rangle \quad \text{or} \quad y \in \langle p \rangle$$

if $x \in \langle p \rangle$ then $\exists n \in \mathbb{Z}$ s.t.

$$x = p^n$$

$$x \cdot y = p^l \Rightarrow p^n y = p^l$$

Note that p has closure so

$$y \in \langle p \rangle.$$

This is a contradiction since $x, y \in (R \setminus \langle p \rangle)$

here, $x \cdot y \in (R \setminus \langle p \rangle)$. $\square$

* Given $x, y \in R$ we call $d \in R$ their Greatest

Common Divisor (gcd)

$$d = \gcd(x, y)$$

if $d|x \wedge d|y$ and if

$$c|x \text{ and } c|y \Rightarrow c|d$$

• Since R is a domain its easy to see $\gcd(x, y)$ is unique up to a factor.

Thm: Let $a, b \in R$. then $d = \gcd(a, b)$ exists and is unique. Moreover,

$\exists x, y \in \mathbb{Z}$ s.t.

$$d = xa + yb.$$

Proof: Define

$$S = \{d \in \mathbb{R} \mid d = xa + yb, x, y \in \mathbb{R}\}$$

$S \subset \mathbb{R}$ which contains a, b since

$$a = 1 \cdot a + 0 \cdot b \in S$$

$$b = 0 \cdot a + 1 \cdot b \in S$$

Since $\mathbb{R}$ is a domain, then all additive inverses exist in S

$$(-a) = (-1)a + 0 \cdot b$$

$$(-b) = 0 \cdot a + (-1)b$$

By the Well-Ordering Principle, S has a smallest element, call it d .

$$\inf(S) = d$$

$d \in S$, so $\exists x, y \in \mathbb{R}$ s.t.

$$d = xa + yb$$

... Use division algorithm.. -

* P is irreducible if, whenever $p = xy$, either x or y is a unit.

• If a nonzero nonunit p is not irreducible then $p = x_1 y_1$, where x_1 and y_1 are non units. Continue until it is a product of irreducible factors.

If the process stops we say the factoring terminates

• We call R a Unique Factoring Domain (UFD)

or a Factorial Ring if

(i) Factoring in R terminates

(ii) Any factorization is unique up to order and units.

• (i) implies every element is a finite product

of irreducibles. The converse is false.

- (i) also holds iff every ascending chain of principal ideals

$$\langle x_1 \rangle \subset \langle x_2 \rangle \subset \dots \text{ stabilizes}$$

If (i) holds, then (ii) holds iff every irred. element is prime.

ex; of UFD's: all $\mathbb{F}$, $\mathbb{Z}$, a polynomial ring L in n variables over a UFD

Lemma: Let $\varphi: R \rightarrow R'$ be a ring map. and $T \subset R'$ a subset. If T is multiplicative, then $\varphi^{-1}(T)$ is multiplicative; the converse holds if φ is surjective

Proof: $\varphi: R \rightarrow R'$

- Let $S \equiv \varphi^{-1}(T)$

If T is mult. then $1' \in T$

$$\text{then } \varphi(1) = 1'$$

$$\Rightarrow 1 \in S$$

Let $x, y \in S$, then

$$x \cdot y \mapsto \varphi(x \cdot y) = \varphi(x) \varphi(y) \in T$$

$$\text{thus } x \cdot y \in S$$

$\therefore S$ is multiplicative.

- If S is multiplicative, then

$$\varphi(1) = 1'$$

$$\text{so } 1' \in T$$

$$x, y \in S \Rightarrow x \cdot y \in S$$

$$\varphi(x), \varphi(y), \varphi(x \cdot y) \in T$$

If φ is surjective, then every $x' \in T$ is of the form

$$x' = \varphi(x), x \in S$$

Thus,

$$\varphi(x \cdot y) = \varphi(x) \cdot \varphi(y) \in T$$

So if φ is surjective, then T is mult. if $\varphi^{-1}T$ is.

Prop: Let $\varphi: R \rightarrow R'$ be a ring map, $q \subset R'$ an ideal. Set $P := \varphi^{-1}(q)$. If q is prime, then P is prime; the converse holds if φ is surjective.

proof: $(R' \setminus q)$ is mult. then $\varphi^{-1}(R' \setminus q)$

is multiplicative. But note $(R \setminus \varphi^{-1}(R' \setminus q)) \cong P$

and so P is prime.

Corollary: Let R be a ring, P is an ideal. P is prime iff R/P is a domain.

proof: $0 \in P$. P is prime iff $\{0\} \subset R/P$ is. $(R \setminus P)$ is multiplicative. Further, $0 \notin (R \setminus P)$ and so $R \setminus P$ is a domain.

$$R/P = \{x + P \mid x \in R\}$$

$$R \xrightarrow{K} R/P$$

K : is a ring map, which is surjective.

$$R \xrightarrow{K} R/P$$

$\cap$

$$P \xrightarrow{K} \{0\} \subset R/P$$

$\{0\}$ is an ideal in R/P

• Now, from above P is prime iff

$\{0\}$ is prime in R/P .

Which $(R/P = \{0\})$ is mult. So R/P is a domain.

Exercise: (2.9) $P := R[X, Y]$. Take the ring map: $\varphi: P \rightarrow A$, $\varphi := \langle X \rangle$

- Surjective

P is prime iff P/φ is a domain from above

- How is φ an ideal of P ?.....

let $\alpha \in \varphi$ and $\beta \in P$, then

$$\alpha \cdot \beta \notin \varphi \dots$$

Not sure how this occurs...

$$P/\varphi \cong R[Y]$$

$R[Y]$ is a domain if R is.

Def: Let R be a ring. An ideal $\mathfrak{m}$ is said to be maximal if $\mathfrak{m}$ is proper and if $\nexists$ ^{there is} proper ideal $\mathfrak{a}$ with $\mathfrak{m} \subsetneq \mathfrak{a}$.

- Let R be a domain. $R[X, Y]$ the polyn. ring. Then $\langle X \rangle$ is prime. However $\langle X \rangle$ is not maximal since $\langle X \rangle \subsetneq \langle X, Y \rangle$.
 $\langle X, Y \rangle$ is maximal iff R is a field.

Prop: A ring R is a field iff $\langle 0 \rangle$ is a maximal ideal.

Proof: Let R be a field. let $\mathfrak{a}$ be a nonzero ideal and a a nonzero element of $\mathfrak{a}$. Since R is a field, $a \in R^\times$, i.e. $a \neq 0$.

$$\text{But } R^\times \cup \{0\} = R$$

and so $\langle 0 \rangle$ is maximal.

• $\Delta = R$ iff $1 \in \Delta$.
 If $1 \in \Delta$, then
 $x = x \cdot 1 \in \Delta \quad \forall x \in R$

$\Delta = R$ iff Δ contains a unit

→ If $\langle x \rangle = R$, x is a unit since $\exists y$

s.t. $xy = 1$

$x \in R$ and $1 \in R$

so $xy = 1$

• Suppose R is a field: let Δ be a nonzero ideal $\Delta \neq 0$. $\alpha \neq 0 \in \Delta$

→ Since R is a field $R \setminus \{0\} = R^\times$
 $\alpha \in R^\times$

Thus, $\Delta = R$.

($\Leftarrow$) Suppose $\langle 0 \rangle$ is maximal. Take $x \neq 0$
 then $\langle x \rangle \neq \langle 0 \rangle$

$\langle x \rangle = R$

x is a unit and so R is a field.

Corollary: Let R be a ring, M an ideal. Then
 M is maximal iff R/M is a field.

Proof:
$$\begin{array}{ccc} R & \xrightarrow{\kappa} & R/M \\ \cup & & \cup \\ M & \xrightarrow{\kappa} & [0] \end{array}$$

• M is maximal in R iff $[0]$ is maximal in R/M . Thus R/M is a field.

R is prime $\Leftrightarrow R/P$ is an integral domain

M is maximal $\Leftrightarrow R/M$ is a field

